



SHRI VILE PARLE KELAVANI MANDAL'S
SHRI BHAGUBHAI MAFATLAL POLYTECHNIC



Computer Engineering Department

SELF LEARNING ASSESSMENT

Submitted in partial fulfillment of the requirements of the
DIPLOMA course by:

Project Topic: EMI Minder	
ROLL NO: B022	
SAPID: 57507240055	
COURSE	Event Driven & UI Programming (EUP230804)
SEMESTER	3 rd DIV B
ACADEMICYEAR	2025-2026

THIS IS TO CERTIFY THAT

Ms. Simran Yagnesh Joshi,
Has satisfactorily completed her Self Learning Assessment and has
submitted the report

Shri Vile Parle Kelavani Mandal's
SHRI BHAGUBHAI MAFATLAL POLYTECHNIC
Department of Computer Engineering

VISION

Create a sustainable academic environment to produce highly competent computer professionals of the future

MISSION

1. To expose students to latest tools and technologies in computing.
2. To foster the professional development of students by providing excellence in education.
3. To Adapt rapid advancements in computing by engaging students in the lifelong learning.
4. To inculcate sound ethical, moral and social values amongst students for benefit of the society.

Program Educational Objectives (PEO's):

The program educational objectives (PEOs) are to produce Computer Professionals

1. Identify, design and solve computing problems by applying knowledge in Computer engineering.
2. Promote lifelong learning by integrating academic knowledge and practical applications.
3. Depict effective team work and practical skills for holistic development

Program Specific Outcomes (PSO's):

PSO1: Demonstrate the fundamental knowledge in the areas of Operating system, Web Technology, Microprocessor based system and IOT by applying programming skills and developing applications.

PSO2: Administer and manage Open source, Networking, Security and Database domains to enhance student growth.

Programme: Computer Engineering

Semester: III DIV B

Course: Event Driven & UI Programming

Course Code: EUP230804

COURSE OUTCOMES (COs): At the end of the semester student will be able to:

CO No.	COURSE OUTCOME
CO1	Demonstrate familiarity with Integrated development environment.
CO2	Design a user-friendly visual interface using windows forms
CO3	Develop GUI application using classic controls of windows form
CO4	Develop GUI application using custom controls of windows form
CO5	Apply database controls to establish database connectivity
CO6	Use data binding techniques for GUI application

Login Form:

```
using System;
using System.Data.SqlClient;
using System.Windows.Forms;
namespace emi
{
    public partial class Form1 : Form
    {
        string connectionString = @"Data
Source=(LocalDB)\MSSQLLocalDB;AttachDbFilename=C:\Users\roshni\Desktop\eut_practicals\EMI
MINDER\emi\emi\Database1.mdf;Integrated Security=True";

        public Form1()
        {
            InitializeComponent();
        }
        private void button1_Click(object sender, EventArgs e)
        {
            string username = textBox1.Text;
            string password = textBox2.Text;
            SqlConnection conn = new SqlConnection(connectionString);
            conn.Open();
            string query = "SELECT COUNT(*)FROM Users WHERE Username=@username AND
Password=@password";
            SqlCommand cmd=new SqlCommand(query, conn);
            cmd.Parameters.AddWithValue("@username", username);
            cmd.Parameters.AddWithValue ("@password", password);
            int count = (int)cmd.ExecuteScalar();
            conn.Close();
            if (count > 0)
            {
                MessageBox.Show("Login Successful!");
                Form2 frm2=new Form2();
                frm2.Show();
                this.Hide();
            }
            else
            {
                MessageBox.Show("Invalid Username or Password!");
            }
        }

        private void Form1_Load(object sender, EventArgs e)
        {
        }
    }
}
```

Dashboard Form:

```
using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Drawing;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
using System.Windows.Forms;
namespace emi
{
    public partial class Form2 : Form
    {
        public Form2()
        {
            InitializeComponent();
        }

        private void loanToolStripMenuItem_Click(object sender, EventArgs e)
        {
            Form3 frm3= new Form3();
            frm3.ShowDialog();
        }

        private void eMIScheduleToolStripMenuItem1_Click(object sender, EventArgs e)
        {
            Form4 frm4 = new Form4();
            frm4.ShowDialog();
        }

        private void paymentHistoryToolStripMenuItem_Click(object sender, EventArgs e)
        {
            Form5 frm5 = new Form5();
            frm5.ShowDialog();
        }

        private void yESToolStripMenuItem_Click(object sender, EventArgs e)
        {
        }

        private void logoutToolStripMenuItem_Click(object sender, EventArgs e)
        {
        }

        private void yesToolStripMenuItem_Click_1(object sender, EventArgs e)
        {
            this.Hide();
        }

        private void noToolStripMenuItem_Click(object sender, EventArgs e)
        {
            MessageBox.Show("Logout cancelled.");}}}
}
```

```

Loan Form:
using System;
using System.Data.SqlClient;
using System.Runtime.InteropServices.ComTypes;
using System.Windows.Forms;
namespace emi
{
    public partial class Form3:Form
    {
        string connectionString = @"Data
Source=(LocalDB)\MSSQLLocalDB;AttachDbFilename=C:\Users\roshni\Desktop\eup_practicals\EMI
MINDER\emi\emi\Database1.mdf;Integrated Security=True";
        public Form3()
        {
            InitializeComponent();
            comboBox1.Items.Add("Home Loan");
            comboBox1.Items.Add("Personal Loan");
            comboBox1.Items.Add("Car Loan");
        }
        private void button1_Click(object sender, EventArgs e)
        {
            double principal=double.Parse(textBox1.Text);
            double annualRate=double.Parse(textBox2.Text);
            int tenureMonths=int.Parse(textBox3.Text)*12;
            double monthlyRate = (annualRate / 100) / 12;
            double emi = (principal * monthlyRate * Math.Pow(1 + monthlyRate, tenureMonths)) / (Math.Pow(1 +
monthlyRate, tenureMonths) - 1);
            label6.Text= emi.ToString("0.00");
        }
        private void button2_Click(object sender, EventArgs e)
        {
            string LoanType=comboBox1.SelectedItem.ToString();
            double principal = double.Parse(textBox1.Text);
            double annualRate=double.Parse(textBox2.Text);
            int tenureMonths=int.Parse (textBox3.Text);
            DateTime startdate = dateTimePicker1.Value;
            double emi = double.Parse(label6.Text);
            SqlConnection conn = new SqlConnection(connectionString);
            conn.Open();
            string query = "INSERT INTO LOANS (LoanType,Principal,InterestRate,TenureMonths,StartDate,EMI)" +
"VALUES(@LoanType,@Principal,@InterestRate,@TenureMonths,@StartDate,@EMI)";
            SqlCommand cmd=new SqlCommand(query, conn);
            cmd.Parameters.AddWithValue("@LoanType",LoanType);
            cmd.Parameters.AddWithValue("@Principal", principal);
            cmd.Parameters.AddWithValue("@InterestRate", annualRate);
            cmd.Parameters.AddWithValue("@TenureMonths", tenureMonths);
            cmd.Parameters.AddWithValue("@StartDate",startdate);
            cmd.Parameters.AddWithValue("@EMI",emi);
            cmd.ExecuteNonQuery();
            conn.Close();
            MessageBox.Show("Loan Details saved successfully!");
        }

        private void Form3_Load(object sender, EventArgs e)
        {
        }
    }
}

```

EMI Schedule Form:

```
using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Drawing;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
using System.Windows.Forms;

namespace emi
{
    public partial class Form4 : Form
    {
        public Form4()
        {
            InitializeComponent();
            dataGridView1.AutoSizeColumnsMode = DataGridViewAutoSizeColumnsMode.Fill;
        }

        private void button1_Click(object sender, EventArgs e)
        {
            double principal = double.Parse(textBox1.Text);
            double annualRate = double.Parse(textBox2.Text);
            int tenureYears = int.Parse(textBox3.Text);
            DateTime startdate = dateTimePicker1.Value;
            int tenureMonths = tenureYears * 12;
            double monthlyRate = annualRate / 12 / 100;
            double emi = (principal * monthlyRate * Math.Pow(1 +
monthlyRate, tenureMonths)) / (Math.Pow(1 + monthlyRate, tenureMonths) - 1);
            DataTable dt = new DataTable();
            dt.Columns.Add("Month");
            dt.Columns.Add("Installment Date");
            dt.Columns.Add("EMI");
            dt.Columns.Add("Interest");
            dt.Columns.Add("Principal");
            dt.Columns.Add("Balance");
            double balance = principal;
            for (int i = 1; i <= tenureMonths; i++)
            {
                double interest = balance * monthlyRate;
                double principalPart = emi - interest;
                balance -= principalPart;
                dt.Rows.Add
                (
                    i,
                    startdate.AddMonths(i).ToShortDateString(),
                    emi.ToString("F2"),
                    interest.ToString("F2"),
                    principalPart.ToString("F2"),
                    balance > 0 ? balance.ToString("F2") : "0:00"
                );
            }
            dataGridView1.DataSource = dt;
        }
    }
}
```

Payment History Form:

```
using System;
using System.Data;
using System.Data.SqlClient;
using System.Windows.Forms;

namespace emi
{
    public partial class Form5 : Form
    {
        string connectionString = @"Data
Source=(LocalDB)\MSSQLLocalDB;AttachDbFilename=C:\Users\roshni\Desktop\eup_practicals\EMI
MINDER\emi\emi\Database1.mdf;Integrated Security=True";

        public Form5()
        {
            InitializeComponent();
            LoadLoans();
            dataGridView1.AutoSizeColumnsMode = DataGridViewAutoSizeColumnsMode.Fill;
        }

        private void LoadLoans()
        {
            comboBox1.Items.Clear();
            using (SqlConnection conn = new SqlConnection(connectionString))
            {
                conn.Open();
                string query = "SELECT DISTINCT LoanType FROM Loans";
                SqlCommand cmd = new SqlCommand(query, conn);
                SqlDataReader reader = cmd.ExecuteReader();

                while (reader.Read())
                {
                    comboBox1.Items.Add(reader["LoanType"].ToString());
                }

                reader.Close();
            }

            if (comboBox1.Items.Count > 0 && comboBox1.SelectedIndex == -1)
                comboBox1.SelectedIndex = 0;
        }

        public void RefreshLoanComboBox()
        {
            string selectedLoan = comboBox1.SelectedItem?.ToString();
            LoadLoans();
            if (selectedLoan != null && comboBox1.Items.Contains(selectedLoan))
                comboBox1.SelectedItem = selectedLoan;
        }

        private void button1_Click(object sender, EventArgs e)
        {
            LoadPaymentHistory();
        }

        private void LoadPaymentHistory()
        {

```

```

        if (comboBox1.SelectedIndex == -1)
        {
            MessageBox.Show("Please select a loan.");
            return;
        }

        string loanType = comboBox1.SelectedItem.ToString();
        DataTable dt = new DataTable();

        using (SqlConnection conn = new SqlConnection(connectionString))
        {
            conn.Open();
            string query = "SELECT PaymentDate, Amount FROM Payments WHERE LoanType=@LoanType ORDER
BY PaymentDate";
            SqlCommand cmd = new SqlCommand(query, conn);
            cmd.Parameters.AddWithValue("@LoanType", loanType);

            SqlDataAdapter da = new SqlDataAdapter(cmd);
            da.Fill(dt);
        }

        dataGridView1.DataSource = dt;
    }

    private void button2_Click(object sender, EventArgs e)
    {
        if (comboBox1.SelectedIndex == -1 || string.IsNullOrEmpty(textBox1.Text))
        {
            MessageBox.Show("Please select a loan and enter an amount.");
            return;
        }

        string loanType = comboBox1.SelectedItem.ToString();
        if (!double.TryParse(textBox1.Text, out double amount))
        {
            MessageBox.Show("Enter a valid numeric amount.");
            return;
        }

        DateTime paymentDate = dateTimePicker1.Value;

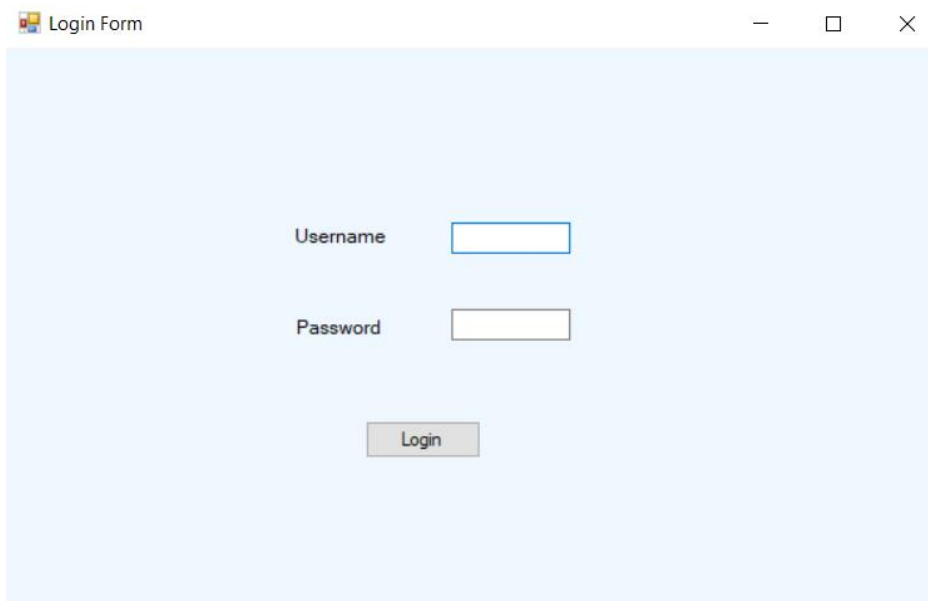
        using (SqlConnection conn = new SqlConnection(connectionString))
        {
            conn.Open();
            string query = "INSERT INTO Payments (LoanType, PaymentDate, Amount) VALUES (@LoanType,
@PaymentDate, @Amount)";
            SqlCommand cmd = new SqlCommand(query, conn);
            cmd.Parameters.AddWithValue("@LoanType", loanType);
            cmd.Parameters.AddWithValue("@PaymentDate", paymentDate);
            cmd.Parameters.AddWithValue("@Amount", amount);
            cmd.ExecuteNonQuery();
        }

        MessageBox.Show("Payment added successfully!");
        textBox1.Clear();
        LoadPaymentHistory();
    }
}

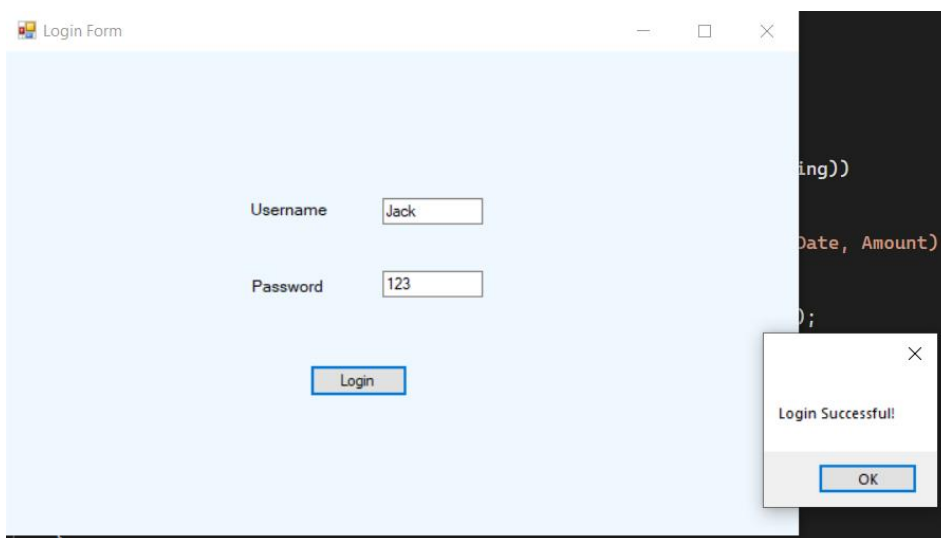
```

Outputs:

Login Form:

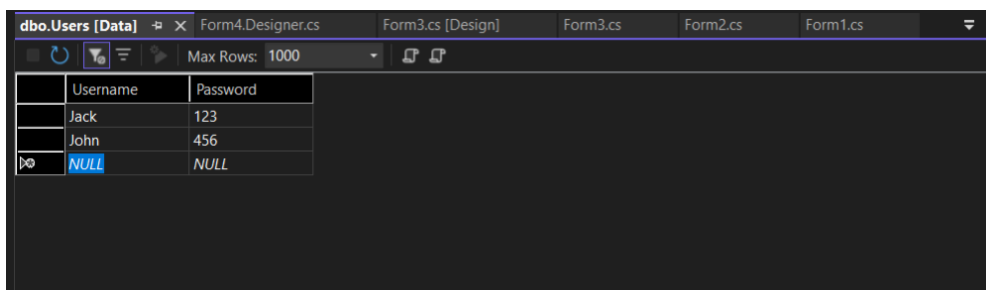
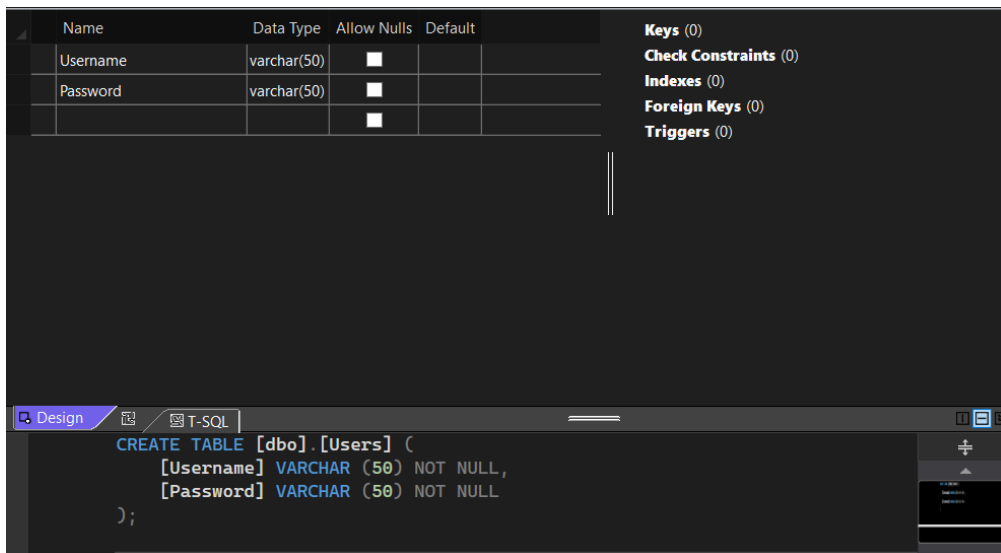


A screenshot of a Windows-style application window titled "Login Form". The window has a light blue background and standard Windows window controls (minimize, maximize, close) in the top right corner. The form contains two labels, "Username" and "Password", each followed by a text input field. Below these fields is a "Login" button. The input fields are currently empty.



A screenshot of the same "Login Form" window, but now with user input. The "Username" field contains the text "Jack" and the "Password" field contains the text "123". The "Login" button is highlighted with a blue border. A small dialog box is open in the bottom right corner of the window, displaying the message "Login Successfull!" (note the typo) and an "OK" button. In the background, behind the login form, some code is visible, including a function definition:

```
ing))  
Date, Amount)  
);
```



Dashboard Form:



Loan Form:

Dashboard Form

Loan

EMI Schedule

Payment History

Logout



Loan Form

Loan Type

Personal Loan ▾

Principal:

1000000

Interest Rate (%):

7

Tanure (Years):

5

Start Date:

02 October 2025 ▾

Calculate EMI

Save Loan

Calculate EMI:

19801.20

Name	Data Type	Allow Nulls	Default
LoanType	varchar(50)	☐	
Principal	decimal(18,2)	☐	
InterestRate	decimal(18,2)	☐	
TenureMonths	int	☐	
StartDate	date	☐	
EMI	decimal(18,2)	☐	

Keys (0)
Check Constraints (0)
Indexes (0)
Foreign Keys (0)
Triggers (0)

```

CREATE TABLE [dbo].[Loans] (
    [LoanType] VARCHAR (50) NOT NULL,
    [Principal] DECIMAL (18, 2) NOT NULL,
    [InterestRate] DECIMAL (18, 2) NOT NULL,
    [TenureMonths] INT NOT NULL,
    [StartDate] DATE NOT NULL,
    [EMI] DECIMAL (18, 2) NOT NULL
);

```

Max Rows: 1000

LoanType	Principal	InterestRate	TenureMonths	StartDate	EMI
Home Loan	1200000.00	3.00	2	11-09-2025	602250.94
Home Loan	10000.00	2.00	3	11-09-2025	286.43
Home Loan	100000.00	2.00	1	11-09-2025	8423.89
Home Loan	100000.00	2.00	10	11-09-2025	920.13
Home Loan	100000.00	10.00	5	11-09-2025	2124.70
Home Loan	100000.00	10.00	5	11-09-2025	2124.70
Home Loan	1000000.00	10.00	5	12-09-2025	21247.04
Home Loan	1000000.00	10.00	5	12-09-2025	21247.04
Home Loan	100000.00	12.00	5	12-09-2025	2224.44
Home Loan	1000.00	1.00	2	14-09-2025	42.10
Personal Loan	1000.00	1.00	2	14-09-2025	42.10
Car Loan	50000.00	3.00	5	21-09-2025	898.43
Home Loan	100000.00	20.00	4	29-09-2025	3043.04
Car Loan	10000.00	10.00	2	29-09-2025	461.45
Personal Loan	100000.00	2.00	3	29-09-2025	2864.26
Personal Loan	100000.00	2.00	4	29-09-2025	2169.51
Personal Loan	1000000.00	7.00	5	02-10-2025	19801.20
Personal Loan	1000000.00	7.00	5	02-10-2025	19801.20
Home Loan	10000000.00	7.00	5	08-10-2025	198011.99
Home Loan	1000000.00	10.00	7	08-10-2025	16601.18
Home Loan	1000000.00	7.00	4	08-10-2025	23946.24
Car Loan	10000.00	4.00	6	09-10-2025	156.45
Home Loan	100000.00	7.00	4	09-10-2025	2394.62
NULL	NULL	NULL	NULL	NULL	NULL

Loan Form

Loan Type

Personal Loan

Principal:

1000000

Interest Rate (%):

7

Tanure (Years):

5

Start Date:

02 October 2025

Calculate EMI

Save Loan

Calculate EMI:

19801.20

EMI MINDER

Form4.cs

Form5.cs

textBox1

);

);

Amount) VALUES

Loan Details saved successfully!

OK

EMI Schedule Form:

Dashboard Form

Loan

EMI Schedule

Payment History

Logout

EMI Schedule Form

Principal:1000000

Annual Interest Rate (%):7

Tenure (Years):5

Start Date:02 October 2025

Generate EMI

Month	Installment Date	EMI	Interest	Principal	Balance
1	02-11-2025	19801.20	5833.33	13967.87	986032.13
2	02-12-2025	19801.20	5781.85	14049.34	971982.79
3	02-01-2026	19801.20	5689.90	14131.30	957851.49
4	02-02-2026	19801.20	5587.47	14213.73	943637.76
5	02-03-2026	19801.20	5504.55	14296.64	929341.11
6	02-04-2026	19801.20	5421.16	14380.04	914961.07
7	02-05-2026	19801.20	5337.27	14463.93	900497.15
8	02-06-2026	19801.20	5252.90	14548.30	885948.85
9	02-07-2026	19801.20	5168.03	14633.16	871315.69
10	02-08-2026	19801.20	5082.67	14718.52	856597.16
11	02-09-2026	19801.20	4996.82	14804.38	841792.78
12	02-10-2026	19801.20	4910.46	14890.74	826902.04
13	02-11-2026	19801.20	4823.60	14977.60	811924.44
14	02-12-2026	19801.20	4736.23	15064.97	796859.46

EMI Schedule Form

Principal:1000000

Annual Interest Rate (%):7

Tenure (Years):5

Start Date:02 October 2025

Generate EMI

Month	Installment Date	EMI	Interest	Principal	Balance
14	02-12-2026	19801.20	4736.23	15064.97	796859.46
15	02-01-2027	19801.20	4648.35	15152.85	781706.61
16	02-02-2027	19801.20	4559.96	15241.24	766465.37
17	02-03-2027	19801.20	4471.05	15330.15	751135.22
18	02-04-2027	19801.20	4381.62	15419.58	735715.64
19	02-05-2027	19801.20	4291.67	15509.52	720206.12
20	02-06-2027	19801.20	4201.20	15600.00	704606.12
21	02-07-2027	19801.20	4110.20	15691.00	688915.12
22	02-08-2027	19801.20	4018.67	15782.53	673132.60
23	02-09-2027	19801.20	3926.61	15874.59	657258.01
24	02-10-2027	19801.20	3834.01	15967.19	641290.81
25	02-11-2027	19801.20	3740.86	16060.34	625230.48
26	02-12-2027	19801.20	3647.18	16154.02	609076.46
27	02-01-2028	19801.20	3552.98	16248.25	592828.20

Payment History Form:

Dashboard Form

Loan EMI Schedule **Payment History** Logout



Payment History Form

Loan Type:

Loan Date:

Loan Amount:

PaymentDate	Amount
14-09-2025	1009.00
02-10-2025	10000000.00

Name	Data Type	Allow Nulls	Default
PaymentID	int	☒	
LoanType	varchar(50)	☐	
PaymentDate	date	☐	
Amount	decimal(18,2)	☐	

Keys (0)
Check Constraints (0)
Indexes (0)
Foreign Keys (0)
Triggers (0)

```

CREATE TABLE [dbo].[Payments] (
    [PaymentID] INT NULL,
    [LoanType] VARCHAR (50) NOT NULL,
    [PaymentDate] DATE NOT NULL,
    [Amount] DECIMAL (18, 2) NOT NULL
);
    
```

	PaymentID	LoanType	PaymentDate	Amount
	NULL	Home Loan	12-09-2025	100000.00
	NULL	Home Loan	12-09-2025	10000.00
	NULL	Home Loan	14-09-2025	1000000.00
	NULL	Personal Loan	14-09-2025	1009.00
	NULL	Car Loan	21-09-2025	1213131.00
	NULL	Car Loan	21-09-2025	100000.00
	NULL	Car Loan	29-09-2025	10000.00
	NULL	Car Loan	29-09-2025	10000.00
	NULL	Car Loan	29-09-2025	100000.00
	NULL	Car Loan	29-09-2025	10000.00
	NULL	Personal Loan	02-10-2025	10000000.00
	NULL	Car Loan	08-10-2025	1000000.00
	NULL	Car Loan	08-10-2025	10000.00
	NULL	Car Loan	09-10-2025	1000.00
	NULL	NULL	NULL	NULL

Server Explorer:

Server Explorer
- Data Connections - Database1.mdf - Tables - Loans - LoanType - Principal - InterestRate - TenureMonths - StartDate - EMI - Payments - PaymentID - LoanType - PaymentDate - Amount - Users - Views - Stored Procedures - Functions - Synonyms - Types - Assemblies - Database2.mdf - Tables - Loans - Views - Stored Procedures - Functions - Synonyms - Types - Assemblies - Servers - DESKTOP-OM0CMDO

Logout:

